

Fig. 3: Actual-size, single-side PCB for the automatic water pump controller

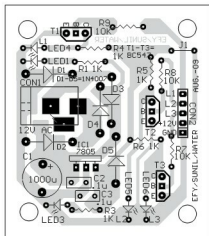


Fig. 4: Component layout for the PCB

plastic switchboard for the ground-floor module, drill holes for six LEDs (four LEDs for level indication, one LED for motor 'on' indication and one LED for power supply indication).

To make the tank module, take a small-size, general-purpose PCB and put all the three cradle switches (CR-S1 through CR-S3) and one four-pin PCB connector following the wiring connections as shown in Fig. 2.

To make the first-floor module, take another small-size, general-purpose PCB, put two four-pin PCB connectors (one for the cable coming from the tank module and the other for the outgoing cable to the ground-floor module) and make appropriate connections with the switches as per Fig. 2. To remove the first-floor module, just remove

switches S1, S3 and S5 with the wires to S3 and S5 left open but wires to S1 closed.

Installation

To protect the tank module from lightening and rain, you can put it inside a soapbox of proper size. Put the tank module circuit inside the soapbox with the help of screws and tighten the balls to cradle switches using appropriate lengths of nylon string as shown in Fig. 1. Fit the first-floor and ground-floor modules on the respective floors and then connect all the three modules using two pairs of telephone wires with the help of four-pin PCB connectors provided on each module.

Testing and working

To test the circuit, close switch S7. LED1 and LED3 glow to indicate that the power supply section is working alright.

Now let us assume that the tank is empty, i.e., water in the tank is below level 1, and therefore all the balls are hanging, meaning that all the cradle switches are 'on.' The high base of all the three transistors drives these into conduction and LED4 through LED6 don't glow. Relay RL1 energises through 12V supply from cradle switch CR-S2 to turn the motor 'on' and LED2 starts glowing.

As water in the tank increases to level 1, cradle switch CR-S1 goes to 'off' condition. As a result, transistor T1 is driven into cut-off state and LED4 glows. Note that the motor is still running as CR-S2 is 'on.'

Now as water reaches level 2, cradle switches CR-S1 and CR-S2 both become 'off' and LED5 also starts glowing. Note that though CR-S2 is 'off,' the motor is still running because cradle switch CR-S3 is 'on,' which provides 12V through S1, S2 and N/O contact of relay RL1.

But when water reaches level 3, cradle switch CR-S3 goes 'off.' As a result, the relay does not get the 12V supply and de-energises to stop the

motor. Now LED6 also starts glowing as cradle switch CR-S3 is 'off.'

The same mechanism happens when water drops from level 3 to level 1. The motor starts automatically when water goes below level 2. So if the power supply is 'on' all the time, the water level is maintained between level 2 and level 3.

Further, let us consider that water in the tank is between level 2 and level 3 and you want to switch on the motor. You can press push-to-on switch S4 or S5 from any floor. The motor will turn off automatically as water reaches level 3. You can also switch off the motor manually by pushing switch S1 or S2 to 'off' position any time.

Now consider that the tank is full and you want fresh water for your fridge bottle. Close switch S3 and then push switch S5 to 'on' position to run the motor from the first floor. For the ground floor, simply close switch S6. Do not forget to open switch S3 or S6 afterwards to resume automatic operation, as these switches are provided for manual operation and will disturb the automatic operation.

In case the automatic circuit fails, open switch S7 and use manual switch S6 to run the motor.

Precautions

1. Connect IC 7812 and transistor BC547 carefully observing their pin configurations, as wrong connection may damage these devices permanently.
2. Ensure proper cable connection from the tank module to the ground-floor module, as any wire interchange will totally disturb the circuit operation.
3. Use a proper heat-sink for IC 7812 as it heats up during operation. In the event of excessive heating, switch off the 12V supply.
4. The transformer also heats up as the circuit is 'on' round the clock. So it is advisable to use a small 12V DC fan inside the ground-floor module. ●

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